



CreatorHub

Production-Grade Brand × Local Creator Collaboration Marketplace

Architecture & Detailed Project Structure Specification • Real Data First Guarantee

Node.js 22 (node:sqlite WAL)

Express 5.2

React 19.2

TypeScript 5.7

Vite 8.2

Tailwind CSS v4.3

Meta Graph API v19.0

Google Gemini GenAI

1. Architectural Overview & Pillars

CreatorHub connects local businesses with authentic, verified Instagram creators within precise geographic radii (1 km – 50 km). Built under a strict "*REAL DATA FIRST. ZERO FAKE DATA*" architectural guarantee, all follower counts, engagement metrics, and media items originate exclusively from official Meta Graph API v19.0 connections or multi-tier search crawlers, with zero synthetic guesswork.

1. Real Data First Guarantee

Official OAuth exchange, long-lived token vault (AES-256), fallback web crawlers, transparent data provenance.

2. Haversine Geospatial Discovery

1km, 5km, 10km, 25km radius filtering with Leaflet interactive OpenStreetMap discovery.

3. Multi-Factor Compatibility Matching

Objective 0–100% score based on distance, engagement compliance, audience tier, and niche overlap.

4. 6-Stage Escrow Lifecycle

APPLIED → ACCEPTED → ESCROW_LOCKED →
PROOF_SUBMITTED → APPROVED →
ESCROW_RELEASED.

2. Master Directory Tree

```
CreatorHub/
├── docs/                                # System documentation, design docs & specifications
│   └── ARCHITECTURE.md                 # System architecture, workflows & scoring formulas
├── public/                             # Static web assets & favicons
├── server/                             # 🟢 Node.js 22 + Express 5 Backend REST API
│   ├── index.cjs                      # Express bootstrap, CORS, JSON parsers & route mounting
│   ├── db/                            # Relational Database Engine
│   │   ├── database.cjs               # node:sqlite WAL connection & PostgreSQL fallback pool
│   │   ├── schema.sql                 # 20 normalized relational tables with foreign keys
│   │   ├── seed.cjs                  # Database seed: Admin, Brands (Third Wave), Creators
│   │   └── migrate_subscriptions.cjs  # Creator Pro subscription tiers schema migration
│   ├── data/                         # Persistent SQLite storage directory (creatorhub.db)
│   ├── middleware/                   # Request Interceptors & Security
│   │   └── auth.cjs                  # JWT token authentication & role guards (Admin/Brand/Creator)
│   ├── models/                      # ORM data-access methods (User, Creator, Brand, Campaign)
│   ├── routes/                      # 15 Express REST API Route Modules
│   │   ├── admin.cjs                 # Platform moderation, user suspension, Meta API health
│   │   ├── ai.cjs                    # Google Gemini AI creator analysis & brief generation
│   │   ├── applications.cjs          # Campaign applications, pitches, status updates
│   │   ├── auth.cjs                  # Sign up, sign in, token generation, role selection
│   │   ├── brands.cjs                # Brand profile management & analytics overview
│   │   ├── campaigns.cjs             # Campaign creation, status filtering, geospatial discovery
│   │   ├── collaborations.cjs        # Deliverables submission, proof review & escrow actions
│   │   ├── creators.cjs              # Creator directory, profiles, direct pitches
│   │   ├── instagram.cjs             # Meta OAuth connect, token refresh & live profile sync
│   │   ├── messages.cjs              # Conversations & direct messaging between brand/creator
│   │   ├── notifications.cjs         # Real-time alert notifications system
│   │   ├── payments.cjs              # Escrow ledger, payout release transactions
│   │   ├── reports.cjs               # Platform safety, content flagging & report logging
│   │   ├── reviews.cjs              # Bidirectional 1-5 star reviews & ratings
│   │   └── subscriptions.cjs         # Creator Pro monetization plans (Free, Silver, Gold, Diamond)
│   └── services/                     # 13 Specialized Business Logic Services
│       ├── AIAnalysisService.cjs      # Google Gemini GenAI prompt creation & structured JSON output
│       ├── InstagramService.cjs       # Meta Graph API v19.0 client & token lifecycle
│       ├── instagramPublicFetcher.cjs  # Multi-tier web crawler (Google, Yahoo, Bing, Jina)
│       ├── MatchingService.cjs        # Haversine distance formula & multi-factor match engine
│       ├── PaymentService.cjs         # Simulated escrow engine & transaction ledger
│       ├── creatorAnalyticsEngine.cjs  # Engagement rates, reach metrics & quality index
│       └── TokenEncryptionService.cjs  # AES-256 encryption for long-lived OAuth access tokens
├── src/                               # 🟢 React 19 + TypeScript + Vite 8 Frontend
│   ├── App.tsx                       # Master React router, role-based redirects & navigation
│   ├── main.tsx                      # React root entry point
│   ├── index.css                     # Tailwind CSS v4 design system tokens & custom fonts
│   ├── context/                      # React State Management
│   │   └── AuthContext.tsx           # Authentication state, login/logout, role session & toast
│   ├── services/                     # API Client Layer
│   │   └── api.ts                    # Typed REST API client communicating with backend
│   ├── types/                        # TypeScript Type Definitions
│   │   └── index.ts                  # Interfaces for User, Campaign, Creator, Collab, Metrics
│   ├── components/                   # Modular UI Components
│   │   ├── layout/                  # Header Navbar & Footer
│   │   ├── instagram/               # Instagram connect, live sync status & handle crawler modal
│   │   ├── analytics/               # AI Creator analysis scorecards & radar metrics
│   │   └── AICampaignModal.tsx       # Gemini AI brief generator popup modal
```

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| └─ CreatorSubscriptionModal.tsx # Silver/Gold/Diamond tier upgrade modal
| └─ DirectPitchModal.tsx        # Direct Brand Pitch & custom budget outreach modal
| └─ ReviewModal.tsx            # Deliverable rating & feedback modal
| └─ pages/                      # 15 Route Views (Dashboards, Feeds, Auth)
|   └─ LandingPage.tsx          # Hero section, feature showcases & live platform stats
|   └─ BrandDashboard.tsx       # Brand control center: campaigns, applicants, escrow
|   └─ CreatorDashboard.tsx     # Creator workspace: briefs, deliverables, payouts, sync
|   └─ CreatorCampaignFeedPage.tsx # Leaflet map & Haversine proximity-based campaign feed
|   └─ CampaignDetailPage.tsx   # Detailed brief description, rewards & application form
|   └─ CreatorProfilePage.tsx   # Public creator media kit, verified metrics & AI audit
|   └─ DirectPitchPage.tsx      # AI Match Studio for discovering top compatible creators
|   └─ CollaborationsPage.tsx   # Active partnerships milestone tracker
|   └─ MessagesPage.tsx        # Real-time brand × creator chat and pitch negotiation
|   └─ AdminDashboardPage.tsx   # Platform moderation, user management, Meta health console
|
└─ package.json                 # Project metadata, npm build/dev scripts & dependencies
└─ vite.config.ts               # Vite bundler, proxy & React compiler settings
└─ netlify.toml                 # Netlify frontend deployment configuration
└─ render.yaml                  # Render backend server hosting blueprint
```

3. Frontend Architecture (`src/`)

The client application is built with React 19, TypeScript, Vite 8, and Tailwind CSS v4, delivering a responsive, dark-mode-first aesthetic with rich interactivity.

3.1 Application Views (`src/pages/`)

Page Component	Route Path	Responsibility & Key Features
LandingPage.tsx	/	High-converting landing page with hero banner, feature breakdown, live creator showcase, and testimonials.
AuthRoleSelectPage.tsx	/choose-role	Entry gateway directing users to authenticate as a Brand Partner or an Independent Creator.
BrandAuthPage.tsx	/brand/login, /register	Brand registration wizard (company name, category, headquarters, website, business email) and login.
CreatorAuthPage.tsx	/creator/login, /register	Creator onboarding flow (niche selection, Instagram handle, location coordinates, minimum rate).
AdminAuthPage.tsx	/admin/login	Secure administrator login portal with quick-fill credentials for pre-seeded dev accounts.
BrandDashboard.tsx	/brand/dashboard	Management hub for publishing briefs, reviewing creator applicants, locking escrow, and releasing payouts.
CreatorDashboard.tsx	/creator/dashboard	Comprehensive creator hub: available briefs, submitted applications, deliverables proof submission, earnings ledger.
CreatorCampaignFeedPage.tsx	/creator/feed	Interactive geospatial feed with Leaflet map pins and Haversine radius sliders (1 km to 50 km).
CampaignDetailPage.tsx	/campaigns/:id	Single brief deep-dive: brand details, deliverables list, budget, geographic radius, and application pitch form.
CreatorProfilePage.tsx	/creators/:id	Public media kit with verified Instagram follower metrics, engagement rate, synced media grid, and Gemini AI score.
DirectPitchPage.tsx	/pitch-creators	AI Matchmaking Studio where brands query and rank nearby creators by compatibility scores (0–100%).
CollaborationsPage.tsx	/collaborations	Milestone workflow tracker (Accepted → Content Creation → Proof Submitted → Approved & Paid).
MessagesPage.tsx	/creator/messages, /messages	Real-time chat portal between brands and creators for reviewing direct collaboration pitches and deliverables.
AdminDashboardPage.tsx	/admin/dashboard	Platform moderation panel: user account suspension, dispute mediation, and Meta Graph API sync health console.

3.2 Core UI Modals & Subcomponents (`src/components/`)

InstagramIntegrationView.tsx

Official Meta OAuth connect flow, manual handle verification, multi-engine public crawler fallback, and live media sync status.

AICreatorAnalysisCard.tsx

Visualizes Gemini AI pillar evaluation (Quality, Consistency, Engagement Health, Commercial Viability, and Strengths).

DirectPitchModal.tsx

Allows brands to send customized proposals (custom budget, required deliverables, pitch note) directly to any creator.

CreatorSubscriptionModal.tsx

Interactive pricing modal for creators to upgrade between Free, Silver, Gold, and Diamond tiers to unlock higher-reward briefs.

4. Backend Architecture (`server/`)

The backend is built with Node.js 22 and Express 5, utilizing the built-in `node:sqlite` engine in WAL (Write-Ahead Logging) mode for zero-dependency local execution, with plug-and-play PostgreSQL support.

4.1 REST API Routes (`server/routes/`)

Route Module	Endpoint Prefix	Operations & Security Rules
auth.cjs	/api/auth	User registration (bcrypt 10 rounds), login, JWT signing, profile resolution, and active role switching.
creators.cjs	/api/creators	Creator listing with geospatial filters, media kits, direct pitch dispatch, and profile editing.
brands.cjs	/api/brands	Brand profile management, active campaign summaries, and brand metrics.
campaigns.cjs	/api/campaigns	Campaign publishing, status transitions, Haversine geospatial proximity search, and match calculations.
applications.cjs	/api/applications	Campaign pitch submissions, subscription quota checks, reward ceiling enforcement, and brand status updates.
collaborations.cjs	/api/collaborations	Active collaboration state machine, deliverable proof URL submissions, revision requests, and escrow payouts.
instagram.cjs	/api/instagram	Meta Graph OAuth token exchange, long-lived 60-day token vault, handle validation, and multi-tier crawler sync.
ai.cjs	/api/ai	Gemini AI-powered creator profile scoring and automated campaign brief generation.
messages.cjs	/api/messages	Thread conversations, unread message counts, and direct communication between collaborating parties.
payments.cjs	/api/payments	Escrow transaction ledger, simulated escrow locking, and verified payout records.
subscriptions.cjs	/api/subscriptions	Tier definitions (Silver, Gold, Diamond), monthly application quota verification, and simulated tier upgrades.
admin.cjs	/api/admin	System-wide KPIs, user ban/unban moderation, dispute resolution, and Meta API sync audit logs.

4.2 Core Micro-Services (`server/services/`)

InstagramService.cjs

Official Meta Graph API v19.0 client. Exchanges short-lived to 60-day tokens, queries `/me/media`, and calculates genuine engagement rates.

instagramPublicFetcher.cjs

Robust multi-tier web crawler (Google, Yahoo, Bing, Jina.ai) for live follower and media extraction when Meta OAuth is in development mode.

AIAnalysisService.cjs

Google Gemini GenAI prompt construction. Generates structured JSON scoring (0–100), key strengths, and monetization advice.

MatchingService.cjs

Calculates spherical geodesic distance using the Haversine formula and computes 6-factor compatibility scores between brands and creators.

PaymentService.cjs

Abstracted escrow engine. Implements *Development Escrow Simulator* with immutable transaction references and audit logs.

TokenEncryptionService.cjs

AES-256-GCM symmetric encryption for securely storing sensitive Meta Graph API OAuth user tokens in the database.

5. Relational Database Architecture (`server/db/`)

The persistence layer consists of **20 normalized relational tables** enforcing strict foreign keys, cascade rules, and timestamp tracking:

Table Name	Primary & Foreign Keys	Domain & Purpose
users	id (PK), email (UNIQUE)	Authentication credentials, password hashes, role designation (creator, brand, admin).
creator_profiles	id (PK), user_id (FK -> users)	Bio, content pillars, coordinates (lat/lng), subscription tier, average rating, rate card.
brand_profiles	id (PK), user_id (FK -> users)	Company name, category, headquarters, store locations, website URL, verified status.
instagram_accounts	id (PK), creator_id (FK)	Meta OAuth tokens (AES-256 encrypted), token expiration timestamp, Instagram Business User ID.
instagram_metrics	id (PK), instagram_account_id (FK)	Verified live follower counts, following, post counts, calculated engagement rates.
instagram_media	id (PK), instagram_account_id (FK)	Synced recent media posts, captions, like/comment counts, media URLs, timestamps.
ai_creator_analyses	id (PK), creator_id (FK)	Gemini AI structured assessment JSON, pillar scores, consistency analysis, monetization tips.
campaigns	id (PK), brand_id (FK -> brand_profiles)	Title, reward per creator, geographic coordinates, target radius, deliverables JSON, status.
campaign_applications	id (PK), campaign_id (FK), creator_id (FK)	Creator pitch message, proposed rate, match score, status (PENDING, SHORTLISTED, ACCEPTED).
collaborations	id (PK), application_id (FK)	Active partnership state machine, current step (1 to 4), delivery deadlines, completion flags.
deliverables	id (PK), collaboration_id (FK)	Proof URLs (Instagram Reel/Story links), submission notes, brand approval/revision status.
payments	id (PK), collaboration_id (FK)	Simulated escrow ledger, held/released status, amount, transaction reference codes.
conversations	id (PK), brand_id (FK), creator_id (FK)	Direct communication channels between brands and creators for pitches and negotiation.
messages	id (PK), conversation_id (FK)	Direct chat messages with timestamp ordering and read status flags.
reviews	id (PK), collaboration_id (FK)	Bidirectional 1–5 star ratings and written performance feedback.
notifications	id (PK), user_id (FK -> users)	In-app notifications for direct pitches, escrow movements, proof reviews, and messages.
admin_actions	id (PK), admin_id (FK -> users)	Audit log of platform moderation (account suspensions, dispute resolutions).

Table Name	Primary & Foreign Keys	Domain & Purpose
refresh_logs	id (PK), account_id (FK)	Meta Graph API sync attempts, duration, error codes, and audit records.

6. Pre-Seeded Demonstration Accounts

Role	Email	Password	Details
Administrator	admin@creatorhub.com	Admin@123	Full governance, user management, and Meta API health console.
Brand Partner	contact@thirdwave.in	Brand@123	Third Wave Coffee (Indiranagar, active campaigns & applicants).
Brand Partner	hello@bluetokai.com	Brand@123	Blue Tokai Coffee Roasters (Koramangala specialty coffee).
Creator	ananya@lifestyle.com	Creator@123	Bengaluru lifestyle & food micro-creator with connected stats.
Creator	rohit@fitness.com	Creator@123	High-engagement fitness creator in Bengaluru.